CATERPILLAR®

DEALER BEST PRACTICE SERIES

Mining Equipment Operation Improvement Program

Maintenance and Repair | **Application** | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

The following document will describe the steps taken by Marcosa to improve the operation of Cat® equipment on a customer site. Marcosa has developed a program that aims to increase operational performance, develop operator skills and knowledge, and reduce overall cost.

2.0 Best Practice Description

The following Best Practice provides details of an operational improvement program developed by Marcosa for a Cat machine fleet. This quarterly program aims to increase production through efficient daily inspections, on the job training and improved motivation of the operators.

During the program, the equipment is monitored for the following criteria: productivity, cost per hour due to operational failure, tires lost due to operational incident, Operator induced events (VIMS and overloads), physical availability, MTBF and MTTR. The operators are assessed in practical and theoretical criteria through a checklist for each model of equipment.

Both equipment and operators are classified in a range of 0 to 5 stars according to the performance (equipment), knowledge and skills (operators).

Operators are split into teams with a team leader, who in this case was called the “godparents” of the equipment. At the end of the program, the equipment with the highest score wins, and each team member receives Cat prizes according to rank. The final score is calculated by the average stars of the equipment x7 and the average stars of the team x 3, divided by 10:

Logo

CAT Prizes

Equipment Measurement Spreadsheet

Marcosa CAT

- Ver Dados Mensais
- Ver Objetivos
- Ver Pesos
- Ver Resultados

DADOS MENSAIS (JAN/11)

MODELO	CRITÉRIOS / FROTA	PRODUTIVIDADE DA EQUIPE OPERACIONAL (META X REALIZADO)	PNEUS PERDIDOS POR ACIDENTE	CUSTO HORA POR FALHAS OPERACIONAIS	QNT DE EVENTOS (VIMS/PRODUCT LINK) SOBRECARGA ≥ 120%	QNT DE EVENTOS (VIMS/PRODUCT LINK) FALHAS OPERACIONAIS	DF	MTBF	MTTR
777F	6001	100%	0	R$ 0,00	0	0	48,73%	48	48,5
	6002	95%	0	R$ 32,84	2	0	94,88%	201,3	3,4
	6003	104%	1	R$ 92,86	5	5	51,51%	37	33,1
	6004	102%	0	R$ 0,00	6	2	66,30%	62,6	32,8
	6005	99%	0	R$ 0,00	9	3	85,93%	111,4	14,7
	6006	98%	2	R$ 33,45	8	8	92,03%	59,9	2,8
	FROTA	99,67%	3	R$ 159,15	30	18	73,23%	86,7	22,55
785D	6101	109%	0	R$ 35,87	10	8	00,35%	31,6	65,6
	6102	111%	0	R$ 39,89	29	61	92,82%	122,8	4
	6103	110%	0	R$ 35,34	22	41	93,49%	125	1
	6104	98%	0	R$ 35,30	0	0	33,85%	83,7	8,5
	6105	109%	0	R$ 0,00	16	55	90,59%	155,8	0,8
	6106	109%	0	R$ 36,64	19	41	96,00%	215	0,3
	6107	106%	0	R$ 0,00	21	38	88,91%	74,01	7,5
	6108	105%	0	R$ 35,31	24	30	91,30%	207,3	0,4
	6109	109%	0	R$ 36,00	14	17	90,59%	101,5	2

RESULTADO

PNEUS PERDIDOS POR ACIDENTE		CUSTO HORA POR FALHAS OPERACIONAIS		QNT DE EVENTOS (VIMS/PRODUCT LINK)				DF		MTBF		MTTR		★
ESTRELAS	PERCENTUAL DA NOTA C/ PESO	ESTRELAS	PERCENTUAL DA NOTA C/ PESO	SOBRECARGA ≥ 120%		FALHAS OPERACIONAIS		ESTRELAS	PERCENTUAL DA NOTA C/ PESO	ESTRELAS	PERCENTUAL DA NOTA C/ PESO	ESTRELAS	PERCENTUAL DA NOTA C/ PESO	RESULTADO (ESTRELAS)
5	16,67%	5	16,67%	5	11,11%	5	11,11%	0	0,00%	2	2,22%	0	0,00%	4,000
5	16,67%	4	13,33%	0	0,00%	5	11,11%	5	11,11%	5	5,56%	5	5,56%	3,833
0	0,00%	4	13,33%	0	0,00%	4	8,89%	0	0,00%	1	1,11%	0	0,00%	2,278
5	16,67%	5	16,67%	0	0,00%	5	11,11%	0	0,00%	3	3,33%	0	0,00%	3,500
5	16,67%	5	16,67%	0	0,00%	5	11,11%	5	11,11%	5	5,56%	0	0,00%	3,944
0	0,00%	4	13,33%	0	0,00%	4	8,89%	5	11,11%	3	3,33%	5	5,56%	3,000
3,333333333	11,11%	4,5	15,00%	0,8333333	1,85%	4,66666667	10,37%	2,5	5,56%	3,166667	3,52%	1,6666667	1,85%	3,426
5	16,67%	4	13,33%	0	0,00%	4	8,89%	0	0,00%	1	1,11%	0	0,00%	3,111
5	16,67%	4	13,33%	0	0,00%	0	0,00%	5	11,11%	5	5,56%	5	5,56%	3,722
5	16,67%	4	13,33%	0	0,00%	0	0,00%	5	11,11%	5	5,56%	5	5,56%	3,722
5	16,67%	4	13,33%	5	11,11%	5	11,11%	0	0,00%	3	3,33%	2	2,22%	3,778
5	16,67%	5	16,67%	0	0,00%	0	0,00%	5	11,11%	5	5,56%	5	5,56%	3,889
5	16,67%	4	13,33%	0	0,00%	0	0,00%	5	11,11%	5	5,56%	5	5,56%	3,722
5	16,67%	5	16,67%	0	0,00%	0	0,00%	5	11,11%	3	3,33%	3	3,33%	3,667
5	16,67%	4	13,33%	0	0,00%	0	0,00%	5	11,11%	5	5,56%	5	5,56%	3,722
5	16,67%	4	13,33%	0	0,00%	1	2,22%	5	11,11%	4	4,44%	5	5,56%	3,778

Operator Evaluation Checklist

Marcosa CAT

CHECKLIST DE AVALIAÇÃO DOS OPERADORES

CAMINHÃO

MIRABELA MINERAÇÃO DO BRASIL
Mina Santa Rita

SITE: MIRABELA   Nome do Operador: Data: 05/11/2010

Pontos verificados		Pontuação	Real
AVALIAÇÃO COMPORTAMENTAL			
1,1	Organizado	10	10
1,2	Pro-ativo	10	10
1,3	Cuidadoso	10	10
1,4	Relacionado	10	10
1,5	Busca excelência	10	10
SEGURANÇA			
2,1	Usa todos os EPI's obrigatórios:	10	10
2,2	Sobe e desce corretamente do equipamento.	10	10
2,3	Limpa todos os pára-brisa da cabine	10	10
2,4	Realiza os ajustes necessários (controles e acento) antes de operar.	10	10
2,5	Certifica que todo dispositivos de advertência (buzina e alarme de marcha à ré) estão funcionando corretamente.	10	10
2,6	Usa o cinto de segurança durante a operação.	10	10
2,7	Buzina para avisar que vai funcionar o motor.	10	10
2,8	Realiza a inspeção da área para ter certeza que não há ninguém próximo do equipamento antes de sair com a máquina	10	10
2,9	Buzina para avisar que vai sair com a máquina.	10	10
INSPEÇÃO DIÁRIA			
3,1	Inspeciona o cinto de segurança quanto a danos, desgastes, fixação e prazo de validade.	10	10
3,2	Inspeciona embaixo do equipamento em busca de vazamentos, danos estruturais e peças soltas.	10	10
3,3	Verifica se a máquina tem o manual de operação e manutenção dentro da cabine.	10	10
3,4	Realiza a inspeção da máquina pelo lado correto (lado direito do equipamento e dar a volta em sentido horário).	10	10
3,5	Verifica as anotações do livro de bordo.	10	10
3,6	Verifica todos os pneus em busca de cortes , parafusos frouxos, danos as travas de aro, tampa da valvula de enchimento faltando.	10	10
3,7	Inspeciona corretamente toda a lateral direita do equipamento em busca de danos estruturais.	10	10
3,8	Inspeciona corretamente toda a parte traseira do equipamento em busca de danos estruturais.	10	10
3,9	Inspeciona corretamente toda a lateral esquerda do equipamento em busca de danos estruturais.	10	10

Nome do Avaliador:____________________

Nome do Supervisor:____________________

Pontuação Máxima Possível	Max. Score:	600
Pontuação Real	Actual Score:	600
Percentual de Conformidade	%:	100,0%
Avaliação Final por Estrelas	Initial Rating	★★★★★

3.0 Implementation Steps

- Collect past 3 months of fleet data:
  - % productivity
  - Cost per hour due to operational failure (service, parts, miscellaneous)
  - Quantity of tires lost by operational accident (cut, collision, spinning)
  - Quantity of operational failure (level 2 VIMS events, overloads)
  - MTBF or MTBS
  - MTTR
- Establish monthly goals for each fleet (Suggestion: + 5% to 10% from baseline for each criteria, over the previous target)
- Develop an operator evaluation checklist for each equipment model
- Develop a spreadsheet to control data and calculate scores
- Generate a training schedule
- Set up meetings to discuss the project with key customer departments
- Define team roles and responsibilities including:
  - o "Godparents" of each piece of equipment
  - o Individual responsible for reporting equipment data
  - o Individual responsible for managing the equipment measurement spreadsheet
- Publicize the program (Logo, T-shirts, banners, e-mails, etc.);
- Begin the program

4.0 Benefits

- **For the Customer:**

  - ✓ Increased operational safety
  - ✓ Costs reduction with maintenance and accidents
  - ✓ Increased tire life due to better underfoot conditions
  - ✓ Increased availability and reliability of the equipment
  - ✓ Increased useful life of the components
  - ✓ Increased productivity
  - ✓ Higher quality operator inspections
  - ✓ Establishment of a new mining culture: Increased ownership of the equipment and operational excellence of the mining equipment
  - ✓ Operator turnover reduction
  - ✓ Better condition of the haul roads, waste pile and loading areas
  - ✓ Additional dealer support and strengthening of the partnership between dealer and customer
  - ✓ Specific training schedule according to the costumer's needs
  - ✓ Greater sense of responsibility by the operators
  - ✓ Greater development of the costumer's manpower
  - ✓ Introduction of new practices, skills and knowledge

- **For the Operator:**
  - ✓ Increased safety
  - ✓ Greater professional development and recognition
  - ✓ Greater sense of team (Responsibility for the equipment)
  - ✓ Greater incentive for self development
  - ✓ Transfer of the knowledge and skills among the operators

- **For the Dealer:**
  - ✓ Greater profitability of the business of their costumer
  - ✓ Greater synergy with the costumer
  - ✓ Be recognized by the costumer as a solution provider
  - ✓ Dissemination of the Cat Best Practices
  - ✓ Improvements in the operation of the costumer's mining equipment
  - ✓ Additional dealer support and strengthening of the partnership between dealer and costumer;
  - ✓ 10% of the realized cost reduction related to maintenance, tires, accidents, etc. will be paid to the Dealer

5.0 Resources Required

- **Human:**

- Continuous Improvement Champion
- Operational Instructor focused on the program.
- Person to gather data and transfer to the spreadsheet;

- **Features:**

- Operator Evaluation Checklist;
- Equipment Measurement spreadsheet;
- Cat Prizes ($7,000 USD)
- Communication material (T-shirts, banners, etc. – $1,800USD)

Total Investments: $8,800 USD

Note: The customer bears the costs with the program.

6.0 Supporting Attachments / References

- Equipment Measurement Spreadsheet
- Operator Evaluation Checklist
- MMM Presentation
- Operator Induced Events.ppt

7.0 Related Best Practices

None

8.0 Acknowledgements

This Best Practice was written by:

Name: Pablo Felipe Lisboa
Title: Continuous Improvement Champion
E-mail: pablo.felipe@marcosa.com.br
Phone: (+55) 71 2107-7574 / 8111-1902